Machine learning methods for continuous glucose monitoring data and their potential in cardiovascular risk assessment

PhD dissertation

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Notes and disclosures

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Notes on website version

This dissertation is available for viewing online at aastedet.github.io/dissertation/ (or by scanning the QR code in the lower right-hand corner of the title page). While the core contents are identical, the website may offer a more comfortable reading experience compared to the physical booklet for some readers. In particular, the website is recommended for readers with impaired vision and others who may be uncomfortable reading the small physical book.

Acknowledgements

Looking back on the last three years against a backdrop of pandemics and full-scale invasions, my time as a PhD student appears minuscule and seems to have gone by in an instant. As it draws to an end, I feel truly grateful for the opportunity to engage in a research topic that was close to me, and for the creative freedom to pursue it in a way that was interesting to me as a researcher and person. This was in large part due to the great people around me during this time, who I owe my thanks.

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This dissertation was built with free, open-source tools, and I would like to extend my gratitude to the many unsung heroes of open-source software development. In particular, to the *R Core Team* for the *R* scripting language and to *Posit* for the *RStudio IDE* and *Quarto* tools.

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Thank you.

Preface

Motivation

During my work as a junior doctor in a general practice in Gellerup, a deprived suburb of Aarhus with a large migrant population, I was immediately struck by the high frequency of type 2 diabetes (T2D) and other chronic diseases among day-to-day consultations. I would soon learn that T2D held more surprises for me than just a high prevalence. After a few weeks, it seemed that much of what I had learned about the disease in medical school and studied in clinical guidelines did not apply in my new setting. Diabetes was not only more prevalent; the patients were also much younger, and their haemoglobin-A1c levels rarely came close to guideline targets despite our best efforts to intensify treatment. Sadly, these experiences would repeat themselves in my subsequent work with migrants as a general practitioner (GP) in health clinics of the Danish Red Cross at asylum residence centres and pre-removal detention centres.

As I reviewed the literature, I understood why medical school and guidelines had failed to prepare me for the challenges I faced as a clinician with T2D in migrants; the existing literature was inadequate. While increased prevalence of T2D appeared well-established, and migrants with T2D were also ascribed a higher mortality than their native counterparts, this provided little guidance for my day-to-day work. Contrarily, there was hardly any evidence on disparities in the time between diagnosis and death - the time when care from the GP is needed the most. So I was left to wonder what might cause this discordance between what I knew about T2D from books and guidelines, and what I encountered in migrants in the clinic.

Are migrants with T2D more prone to under-treatment, and in which areas of care?
Are some migrant groups more prone to under-treatment than others?

Knowledge on these clinical questions could enable GPs and healthcare planners to address disparities and improve care in migrants by prioritising and focussing care accordingly. While migrants are currently a younger demographic group than the rest of the population, it is likely that migrants will constitute an increasing proportion of the T2D population in the coming years as the group ages. Therefore, I was excited to explore these clinical questions as a researcher, hoping I could provide answers to myself and fellow GPs that could lead to better care - and, ultimately, better health - in a vulnerable and challenging group of patients.

During my time as a PhD student, I quickly encountered the first of many challenges on the way to studying migrants with T2D in the Danish registers: there was no validated definition of T2D, nor a common consensus among researchers. This led to an expansion of the scope of the PhD project to develop and validate a tool to define T2D in the Danish registers, which allowed me to answer my research questions based on robust findings. This PhD project benefited greatly from open-source tools. In the spirit of open-source, the source code of the validated diabetes classifier was made available to other researchers in the *osdc* package for the *R* statistical programming language. As a final commitment to openness and accessibility, this dissertation was made available to a global public audience in website format.

Outline of the dissertation

Chapter 1 describes type 2 diabetes (T2D) and how migrants are at particular risk. It then outlines the context of migration and health care in Denmark and clinical guideline recommendations for T2D. Finally, it introduces the reader to identification of diabetes patients in Danish healthcare registers.

Chapter 2 states the overall aims of this PhD project and each individual study.

?@sec-methods describes the setting, data sources, methods and study designs used in the studies of this PhD project.

?@sec-results presents the main results of the studies.

?@sec-discussion-methods contains a discussion of the methods used and their potential impact on results.

?@sec-discussion-results discusses the main findings of each study and compares them to existing literature. Finally, the clinical implications of the results are discussed.

?@sec-conclusion presents the main conclusions.

?@sec-perspectives addresses the future perspectives raised by the dissertation and discusses the opportunities for research in the evolving research field.

The rest of the dissertation includes a list of **references**, English and Danish **summaries**, a **supplementary** validation analysis, and the three **papers** of the dissertation and their supplementary materials.

Papers in the dissertation

Study I

Isaksen AA, Sandbæk A, Bjerg L. Validation of Register-Based Diabetes Classifiers in Danish Data. *Clinical Epidemiology.* 2023;15:569-581. https://doi.org/10.2147/CLEP.S407019

Study II

Isaksen AA, Sandbæk A, Skriver MV, Andersen GS, Bjerg L (2023) Guideline-level monitoring, biomarker levels and pharmacological treatment in migrants and native Danes with type 2 diabetes: Population-wide analyses. PLOS Global Public Health 3(10): e0001277 https://doi.org/10.1371/journal.pgph.0001277

Study III

Isaksen, AA, Sandbæk, A, Skriver, MV, Bjerg, L. Glucose-lowering drug use in migrants and native Danes with type 2 diabetes: Disparities in combination therapy and drug types. *Diabetes Obesity and Metabolism.* 2023; 1-10. https://doi.org/10.1111/dom.15230

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Abbreviations

ACEI: Angiotensin-converting enzyme-inhibitors
APT: Antiplatelet therapy
ARB: Angiotensin receptor blockers
ATC: Anatomical Therapeutic Chemical (classification)
CI: Confidence interval
CVD: Cardiovascular disease
DKD: Diabetic kidney disease
DPP4i: Dipeptidyl peptidase-4 inhibitors
GP: General practitioner
GDM: Gestational diabetes mellitus
GLD: Glucose-lowering drugs
GLP1RA: Glucagon-like peptide-1 receptor agonists
HbA1c: Haemoglobin-A1c
LDL-C: Low-density lipoprotein cholesterol
LLD: Lipid-lowering drugs
NPV: Negative predictive value
OR: Odds ratio
OSDC: Open-Source Diabetes Classifier
PCOS: Polycystic ovary syndrome
PPV: Positive predictive value
RR: Relative risk
RSCD: Register of Selected Chronic Diseases
SGLT2i:: Sodium glucose co-transporter type 2 inhibitors
T1D: Type 1 diabetes mellitus
T2D: Type 2 diabetes mellitus
UACR: Urine albumin-to-creatinine ratio

1 Introduction

1.1. Diabetes and cardiovascular disease

- Burden of diabetes
  - Type 1 diabetes
  - Type 2 diabetes
- Cardiovascular complications
- Heart failure in diabetes
- NT-proBNP as a biomarker

1.2. Glycemic markers and glucose monitoring

- HbA1c and its limitations
- Continuous glucose monitoring (CGM)
- Glycemic variability
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1.3. Cardiovascular risk prediction in diabetes

- Traditional risk models
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- Data-driven vs hypothesis-drivenly(knowledge-based) paradigms
- Why machine learning in diabetes and cardiovascular risk prediction?
- Why are ML fantatic for CGM data?
- Types of machine learnin g methods
  - Logistic regression
  - LSTM and CNNs
  - Foundation models

- Challenges and pitfalls of machine learning in healthcare
  - Overfitting and generalizability
  - Interpretability and explainability

1.5. Fairness and bias in machine learning

- Define race and ethnicity
- Algorithmic bias and group fairness
- Fairness metrics and mitigation strategies
- Challenges in healthcare applications when using ML

2 Aims

2.1. Overall aims

The overall aim of this PhD dissertation is to provide robust knowledge on specific areas of T2D care where disparities between migrant groups and native Danes may be present.

2.2. Specific aims

I. To develop a register-based classification of type 1 and type 2 diabetes that incorporates previous recommendations on data sources, and to make it available to other researchers.

II. To validate register-based diabetes classifiers available to researchers in Denmark.

III. To assess the prevalence of T2D in migrant minorities of Denmark, and to explore care disparities between migrant groups and native Danes in terms of monitoring, biomarker control and pharmacological treatment as recommended in the guidelines.

IV. To evaluate disparities in the quality of glucose-lowering pharmacological treatment between migrants and native Danes in terms of combination therapy and drug types.

2.3. Outputs

Aim I was pursued in the creation of the *Open-Source Diabetes Classifier* (OSDC), made available in the open-source *R* package *osdc*.[osdc?] Aims II, III, and IV were pursued in the respective papers I, II, and III of this dissertation.

Methods 3

In this chapter the three studies included in this PhD will be presented. The data will be described together with the chosen methods of the three studies and the different clinical variables, CGM-derived metrics and relevant cardiovascular outcomes will be explained.

☐ What was done (Description)
☐ Why it was done (Justification)

3.1. Overview of the studies

	Study I	Study II	Study III
Title	Continuous Glucose Monitoring–Derived Metrics and Cardiovascular Risk Among People With Diabetes: Systematic Scoping Review	External evaluation of a foundation model for continuous glucose monitoring data and cardiovascular risk assessment in people with diabetes	Racial disparities in continuous glucose monitoring-based 60-min glucose predictions among people with type 1 diabetes
Study Type	Review	CGM foundation model evaluation study	Algorithmic fairness study
Data Sources	Literature on Embase and MEDLINE	The CANCAN study and the T1D Exchange (T1DX) dataset	T1D Exchange (T1DX) dataset
Population	Studies including people with type 1 diabetes, type 2 diabetes and/or prediabetes	159 people with type 2 diabetes 205 people with type 1 Diabetes	205 people with type 1 Diabetes
Input, predictors and exposures	Search strategy and eligibility criteria	Clinical variables, CGM-derived metrics and continuous CGM data	Continuous CGM data
Outcome, target	All studies that uses CGM data to say something about cardiovascular disease risk and outcomes	Visualisation of foundation model embeddings and elevated NT-proBNP prediction	Next blood glucose value
Analysis	Systematic scoping review, study characterization, and narrative synthesis	Foundation models and logistic regression models	Long short term memory models (LSTM) and Linear mixed effect models (LMEM)

3.2. Study I: A scoping Review

☐ Explain why scoping review and not a systematic review

A scoping review was conducted to investigate current knowledge of CGM-data and its possible connection and predictive power to clinical and subclinical cardiovascular outcomes. The scoping review followed the Manual for Evidence Synthesis (Chapter 10 - Scoping reviews) from the Joanna Briggs Institute and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses extension for scoping review (PRISMA-ScR) guidelines.[1,2]

3.2.1. Concepts and Definitions

The review focused solely on CGM-data for the input to any association or prediction models in the literature. This meant that studies only investigating self-monitored blood glucose with finger-prick measurements or blood sample measurements such as HbA1c were not considered in this review. CVD outcomes were divided into subclinical or clinical For the clinical CVD, the following outcomes were considered: cardiovascular mortality, major adverse cardiovascular events, coronary artery disease, heart failure, stroke, and peripheral artery disease with synonymous terms included (e.g., ischemic heart diseases) and also clinical events (e.g., undergoing coronary artery bypass surgery). Subclinical CVD was grouped into six subcategories: arterial stiffness, flow resistance, arterial wall thickness, arterial wall composition, cardiac and pulse-related measures, and arterial lumen. CVD outcomes did not include broader risk factors nonspecific to cardiovascular risk (e.g., age or sex). Eligibility criterias for the scoping review[3] (Study I) can be found in Table 3.1.

Table 3.1.: Eligibility criteria for the scoping review (Study I)

Inclusion Criteria	Exclusion Criteria
Human clinical studies including participants with prediabetes or any type of diabetes except for gestational diabetes, regardless of definitions Peer-reviewed published original articles (including brief reports). Studies investigating either: a) The association between CGM-derived metrics of glycemic control and cardiovascular risk markers or cardiovascular diseases (prevalent or incident). b) CGM-derived metrics of glycemic control as predictors of cardiovascular risk markers or cardiovascular diseases (prevalent or incident).	Review articles, editorials, case reports, protocols, conference abstracts, and preprints. Animal studies not including any human participants. Studies not including metrics derived from continuous glucose monitoring device data. Studies including CGM-derived metrics as outcomes. Studies not focusing on CVD outcomes according to our definition, as outlined in the "Concepts and Definitions" section. Studies focusing on pregnant women with any form of diabetes including gestational diabetes. Studies where participants were monitored post-surgery or during hospitalization (e.g. Intensive care unit). Language not understood by authors

3.2.2. Search and Selection of Sources

The databases Embase and Medline were searched from inception to March 11, 2025. The seach strategy was tested against 13 articles in the field, found through an initial exploratory search.[4–16]

All identified records were uploaded into EPPI reviewer 6[17] and duplicates were removed. Titles and abstracts were screened by at least two independent reviewers and a meeting was held after ~5% of the titles and abstracts had been screened, to resolve disagreements and ensure consistency between reviewers. Afterwards full-text screening was performed by two independent reviewers and held up against eligibility criteria. Disagreements were resolved through discussion at physical meetings between reviewers.

Backward and forward citation searching was performed on all the included studies after the screening process using citationchaser[18] and the screen process was repeated until no further new studies were identified through the citationchaser.

3.2.3. Data Charting Process and Data Items

Research questions were defined before the screening process and data was extracted based on these research questions (Note 1) and following the requirements from the PRISMA-ScR checklist.[1]

Note 1: Research Questions (Study 1)

1. Is there an association between glycemic control and CVD risk?
2. Can CGM-derived metrics predict CVD risk?
3. What CGM-derived metrics are used in the literature?
4. Which cardiovascular markers and diseases are included as outcomes in the studies?
5. What characterizes study populations (age, sex, ethnicity, geographic location)?
6. What study designs are used (e.g., longitudinal cohort studies, randomized controlled trials, cross-sectional studies)?
7. How was data collected (e.g., clinical trial, epidemiological study, routinely collected data)?
8. What CGM devices were used?
9. What statistical models were used in the studies?
10. Is the data openly available?
11. Is the code openly available?

3.2.4. Synthesis of Results

Study characteristics were synthesized using descriptive statistics and supplemented by a narrative summary. Only the most adjusted models in the main population for each study was considered in this review. This approach was necessary as several studies reported numerous estimates across varying adjustments, different combinations of CGM-derived metrics and CVD outcomes, and/or distinct subgroup stratifications. Given the large volume of literature, studies investigating clinical cardiovascular outcomes were prioritized in the main synthesis. While subclinical markers offer valuable insight into early vascular changes and disease progression, clinical outcomes are more directly relevant to cardiovascular risk prediction and clinical decision-making. The main synthesis of clinical outcomes included both effect estimates and measures of statistical significance whereas the supplementary synthesis on subclinical CVD focused primarily on creating an overview of the chosen CGM-derived metrics and the statistical significance of reported associations. All evaluation metrics mentioned in the studies doing prediction analyses were reported based on what the studies reported.

3.3. Study Populations

Two datasets were used in this PhD (Study II and study III) which are presented in the following sections

3.3.1. CANCAN

The CANCAN dataset was collected in an observational case-control study from three hospital outpatient clinics (Denmark). The aim of the CANCAN Study was to investigate if cardiac autonomic neuropathy (CAN) assessments, CGM and heart failure indicators could be used in risk stratification in people with T2D. Participants were included if they were adults ($<$ 18 years old) and had T2D for at least a year. Exclusion criteria were atrial fibrillation or laser-treated eye disease within the past three months, were pregnant or lactating, had a life-threatening disease (expected survival $<$ 1 year), or had cognitive disabilities rendering them unable to understand and give consent. Data were collected during a clinical examination, during which blood and urine samples were obtained, and a blinded CGM sensor was inserted. Blood samples to obtain NT-proBNP values were collected when the CGM sensors were returned, after the 2-week CGM data collection period had ended (Figure 3.1). In total, 159 participants were incluceded since the rest were missing age, sex, HbA1c, BMI, NT-proBNP, HDL, total cholesterol, or eGFR and these values were needed for the analyses in Study II.

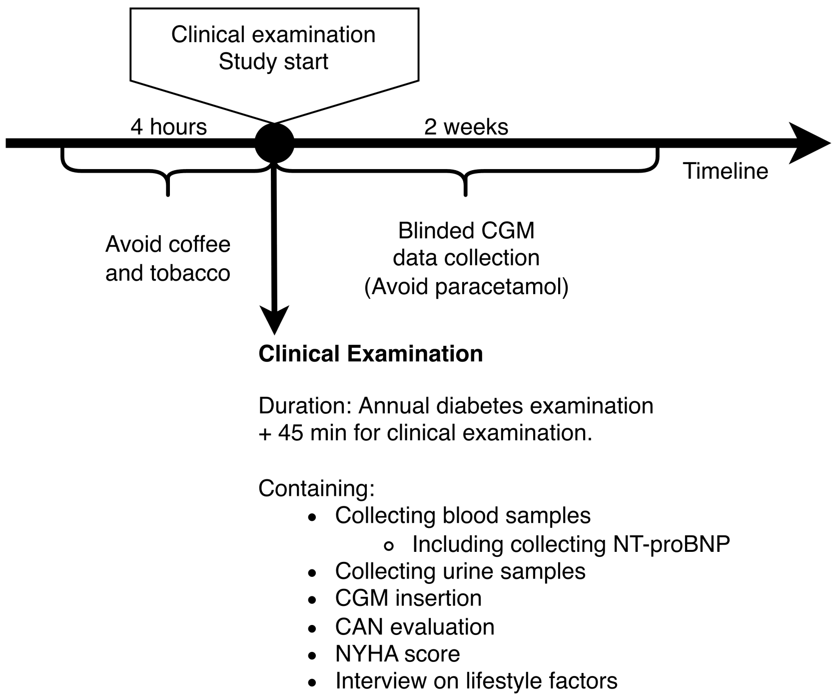


Figure 3.1.: Data collection timeline for the CANCAN Study

3.3.2. T1DX

The publicly available T1D Exchange (T1DX) dataset, distributed by the JAEB Center for Health Research under the protocol "Racial Differences in Mean CGM Glucose in Relation to HbA1c, was collected in 10 diabetes centers (US).[19] The data was collected to determine if there is a racial difference in the association between mean glucose and glycated hemoglobin (HbA1c) between non-Hispanic African Americans or black (Black) and non-Hispanic white (White) people with T1D. In the T1DX study ethnicity and race was defined through self-identified and self-reported ethnicity/race verification forms where ethnicity was defined as Hispanic or Latino as: *"a person of Cuban, Mexican, Puerto Rican, South or Central American, or other Spanish culture or origin, regardless of race. The term "Spanish origin" could be used in addition to "Hispanic" or "Latino" "*.[20] For race a participant defined as White was *"a person having origins in any of the original peoples of Europe, the Middle East or North Africa." and Black or African American as "a person having origins in any of the Black racial groups of Africa. Terms such as "Haitian" or "Negro" can be used in addition to "Black" or "African American" "*.[20] In this thesis, the term race will be used moving forward, as it aligns with the terminology used by Bergenstal et al.[20], it is, however, acknowledge that it is inherently difficult to define race.[21] The T1DX study adhered to the tenets of the Declaration of Helsinki and was approved by the local institutional review boards and 205 people completed the study by providing written informed consent and assent when applicable.[20] All 205 participants were included in further analyses. Of the 205 participants, 101 were White and 104 were Black. For each race, two age groups were sampled 8 to <18 years old and >18. Each participant wore a blinded modified investigational version of Abbott Diabetes Care's Flash Glucose Monitoring System (CGM) as continuously as possible for 12 weeks. CGM data was sampled every 15 minutes in a range between 2.21 - 27.65 mmol/L (40 and 498 mg/dL). Blood samples were taken for measurement at enrollment, at some of the follow-up visits, and at the end of enrollment.

3.4. Study II: Foundation models and predictive utility of CGM data

The second study was focused on investigating the knowledge and predictive utility of CGM data, both extracting a novel CGM-derived metric and the classical known ones found in study I.[3]

3.4.1. Preprocessing of CGM data and CGM-derived metrics

- ☐ days with less then 70 % of data were removed
- ☐ Add table with CGM derived metrics
- ☐ Add CGMformer input information
- ☐ What is SCORE2-Diabetes 10 year CVD risk score?

3.4.2. Foundation Models and Embeddings

- ☐ What is a foundation model
- ☐ What is an embedding
- ☐ What is cosine similarity and what is cosine distances
- ☐ How are the cosine distances used?

3.4.3. Predictive Modelling

- ☐ What is NT-proBNP (scale, unit and connection with heart failure)
- ☐ What is logistic regression and what is regularization techniques and why are they used? (Ridge regression)

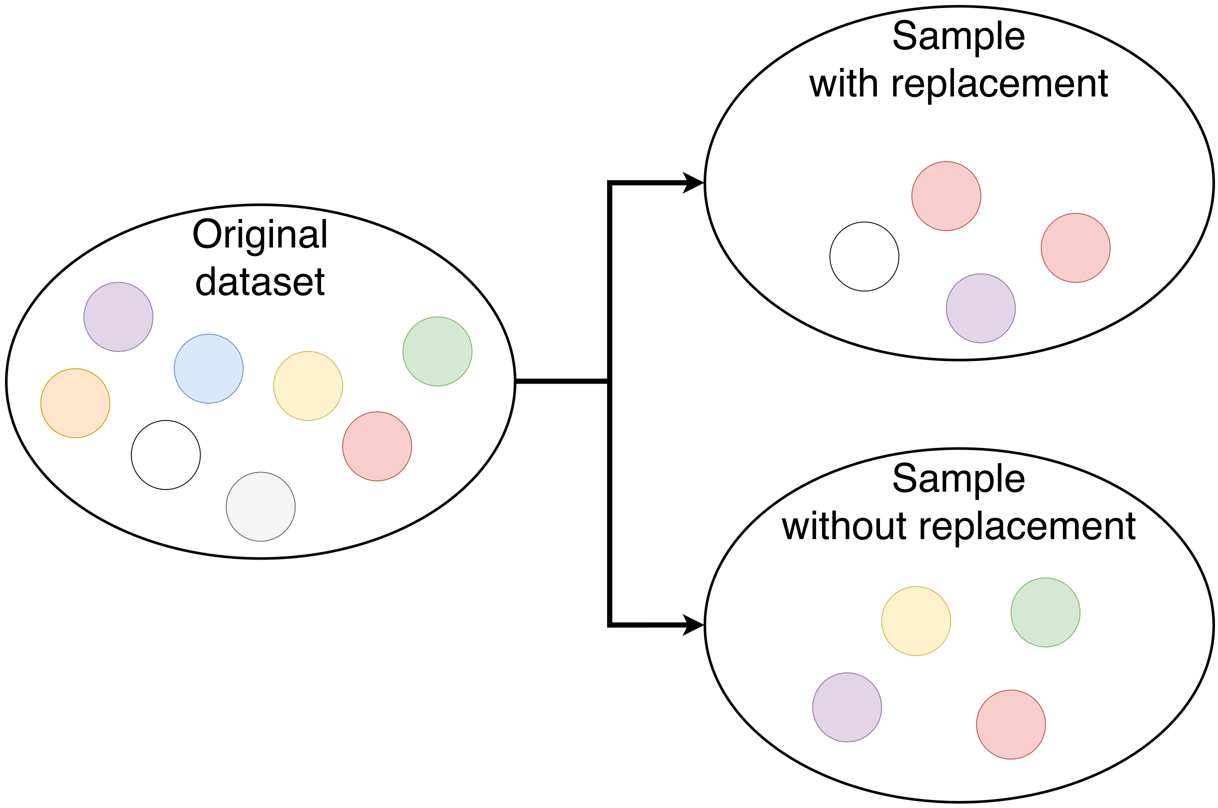


Figure 3.2.: Resampling with and without replacement

3.5. Study III: Algorithmic Fairness in Blood Glucose Forecasting

The third study was investigating racial bias in a data driven machine learning model only based on blood glucose measurements from a CGM-sensor as this is an important consideration in the path to get ML models into use

3.5.1. Preprocessing of CGM data for Forecasting

All data was sorted into 60 minutes windows with 4 values to predict 60-minutes into the future as visualized in Figure 3.3. There are no clear answers in the literature about which prediction horizon is the best to choose, however CGM devices measure in the interstitial fluid and there is a physiological delay between the measured blood glucose and the interstitial glucose levels.**source?** Of the European approved CGM devices it is only the Dexcom that shows trends to the users and since CGM devices mainly focuses on monitoring, the forecasting question is only really interesting when administering insulin in order to catch hypo- and hyperglycemic events ahead of time.**source?** It is very difficult to figure out how the prediction algorithms in insulin pumps are made and so far I have had very little luck in figuring it out. It is assumed that the companies use linear regression and the last 20 minutes of data to predict the next values which is usually the last four samples since the resolution of the used CGM devices usually are between 1 to 5 minutes.**source?** With a CGM measurement every 15 minutes and the physiological delay in mind, the 60 minutes were decided to be a good prediction horizon. All samples with missing values were removed.

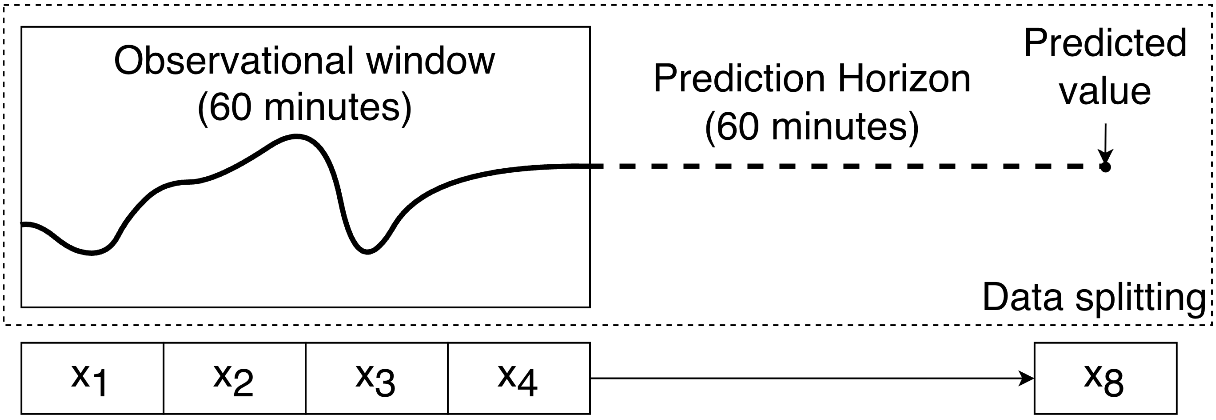


Figure 3.3.: The prediction horizon and overview of the creation of samples (predictors and targets).

3.5.2. Partitioning of the Dataset

The T1DX dataset was resampled without replacement on the participant level to avoid data leakage and to be able to test performance on an individual level. The resampling without replacement is the other method mentioned above in Figure 3.2. It is a form of bootstrapping where a new dataset (subset) is created based on the original dataset by randomly picking samples from the original dataset to create a new subset. Data from each participant was tested on different subsets of data containing varies ratios of Black and White proportions as illustrated in Figure 3.4. By creating 11 different variations (0–100%, 10–90%,..., 100-0%), it meant that 11 subsets were created for each of the participants in the original T1DX dataset. Since some subsets where created with only one race present, it meant that all the subset could only be about half the size of the original T1DX dataset when it came to number of participants. This resulted in 2,255 training datasets (205 individuals times 11 training sets with varying proportions of race) and 205 independent test sets. The subsets were used to train ML models and data leakage was avoided by keeping the train-test-splits at the participant level, meaning that all samples belonging to the same participant were kept together throughout all splits.

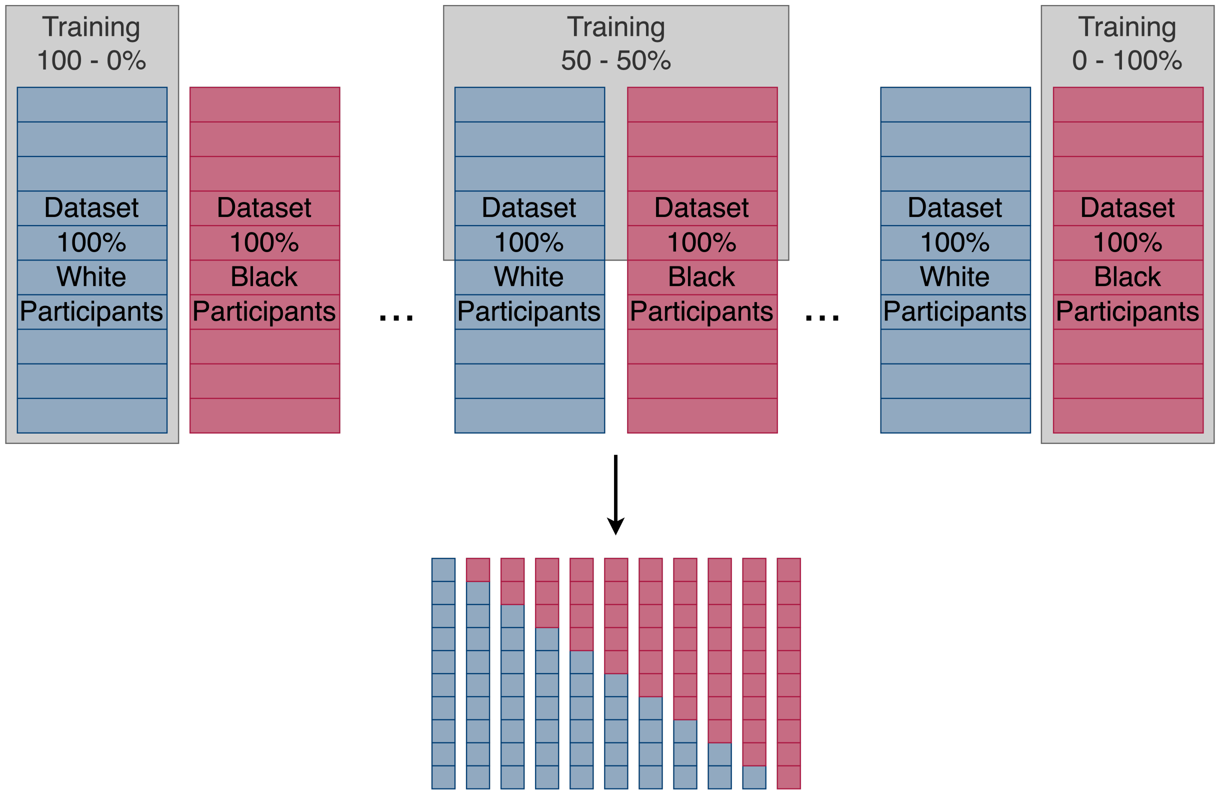


Figure 3.4.: The prediction horizon and overview of the creation of subsets for training.

Eleven subsets were created for each of the participants and the last weeks worth of data was chosen to test on as illustrated in Figure 3.5 The one week was chosen because participants had varying amount of CGM data and the participant with the least amount of data had ~ 2 weeks, and all participants were wished to be kept due to the small number of participants in the T1DX dataset. The remaining data were used for training the models.

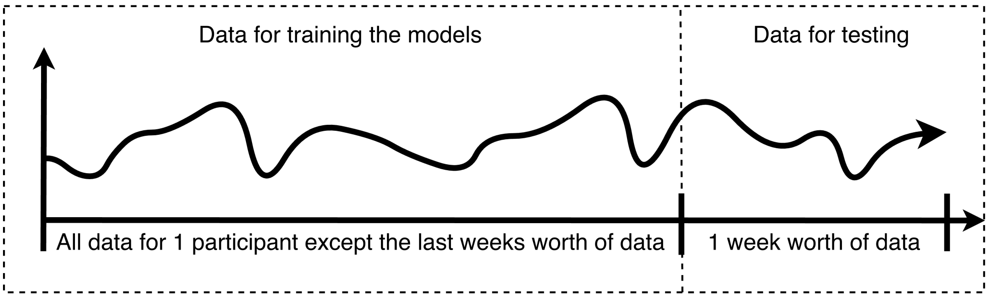


Figure 3.5.: The splitting of a participants data to create training, fine-tuning and test sets

3.5.3. Training Approaches and Model Selection

Four training approaches were developed as illustrated in Figure 3.6. The training approaches included a naive baseline and three different machine learning models as described in Table 3.2.

A long short-term memory model (LSTM) and overall architecture was chosen for the machine learning model, based on a previously described LSTM in van Doorn et al.@vanDoorn2021 (two LSTM layers followed by a dense layer) since there were good performance with this model. All the training data were scaled to be within the range of zero to one. Hyperparameters included the number of neurons in LSTM layers, activation functions, dropout rate, batch size, learning rate, decay rate, and early stopping. These hyperparameters were tuned using a grid search, heatmaps, and manual tuning. Further information is given in in Thomsen et al.[22] and the code repository[23].

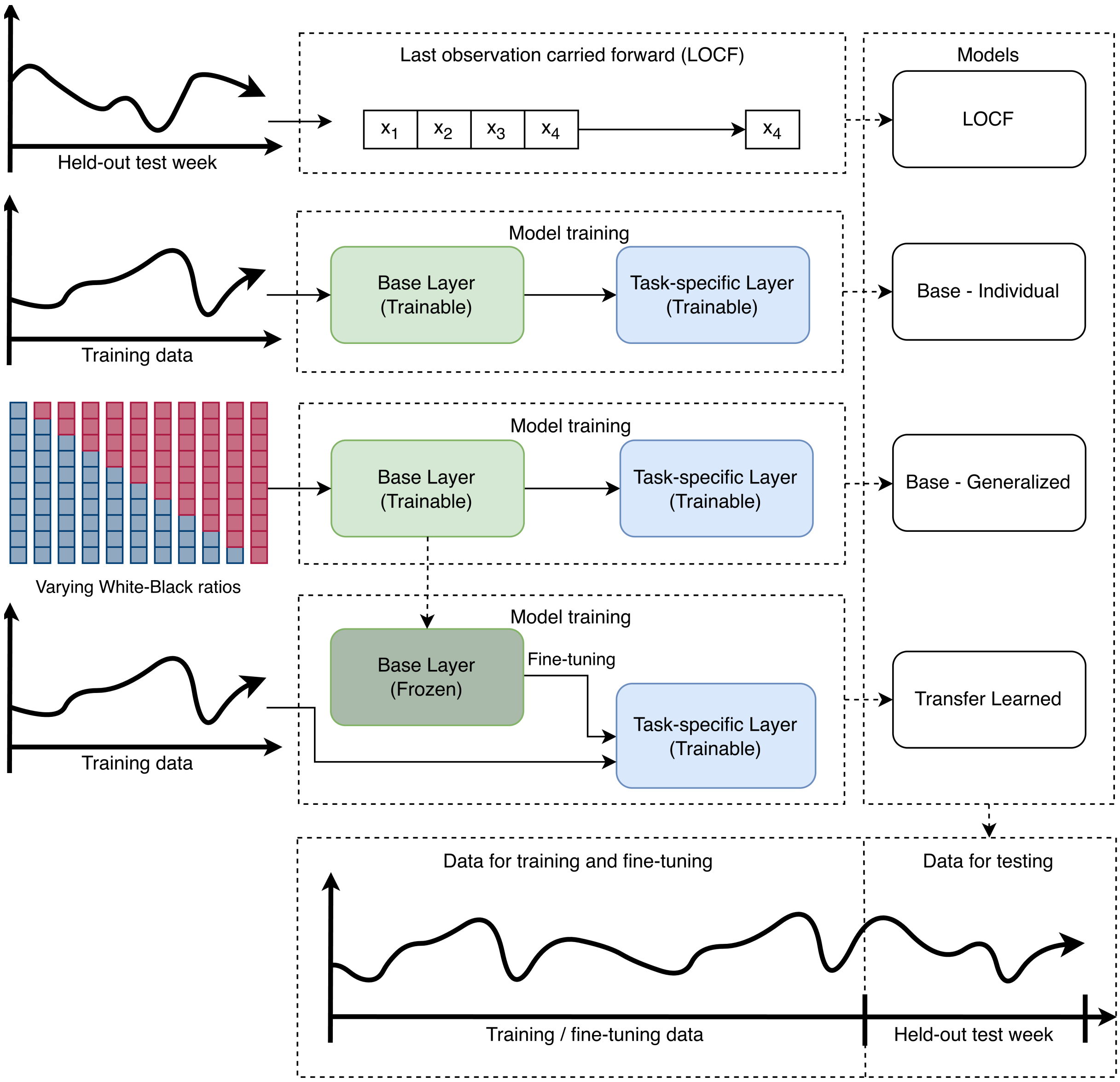


Figure 3.6.: The four training approaches. The Base-Individual, Base-Generalized, and Transfer Learned models were based on an LSTM architecture. The base layer (green) consisted of two LSTM layers, while the task-specific layer (blue) consisted of a single dense layer.

One of the machine learning models were developed using transfer learning. It is a machine learning technique in which knowledge learned from one domain is transferred to another related domain. One common transfer learning approach is fine-tuning, where a pretrained model is further trained using new data.[24] In this analysis, generalized glucose prediction models were first trained using data from multiple participants (Base-Generalized model) and subsequently fine-tuned using participant-specific data to personalize glucose predictions (Transfer Learned model).

Table 3.2.: The four training approaches used to develop glucose prediction models for each participant in the T1DX dataset.

Model names	Description	Training data
Last observation carried forward (LOCF)	naive baseline: In the LOCF model, a given CGM measurement (x4) was used to predict blood glucose values 60 minutes in advance (x8) in the test set by assuming no change in the last 60 minutes.	No training data required
Base-Individual	In the Base-Individual model, each participant's complete CGM data (except for the last week, which was used for the testing data) was used for training.	Only data from 1 participant Figure 3.5
Base-Generalized	Here, for each participant, the 11 training corresponding datasets described above were used to generate 11 separate models, which were all subsequently tested on the given participant's last week of CGM data.	All the different subsets of data Figure 3.4
Transfer Learned	This model mirrored the Base-Generalized Model's setup. However, the 11 trained models for each participant were subsequently tailored, i.e., fine-tuned with transfer learning to the given participant's complete CGM data (except for the last week, which was used for the testing data)	Only data from 1 participant Figure 3.5

3.5.4. Model Evaluation

To evaluate model performance, each participant's last weeks worth of data (672 measurements) was reserved for internal testing (Figure 3.5). Model accuracy was assessed by calculating the root mean squared error (RMSE) for each participant across all four modeling approaches. For the LOCF and Base-Individual models, this yielded one RMSE value for each of the 101 White participants and 104 Black participants. For the Base-Generalized and Transfer Learned models, RMSEs were calculated for each participant across 11 data subsets, each representing a unique proportion of White and Black participants. Performance was summarized as the mean RMSE with corresponding 95% confidence intervals (CIs) for each racial group.

To evaluate the predictive performance across different ratios for the Base-Generalized and Transfer Learned models a linear mixed effect model (LMEM) was used since data was dependent due to participants having an RMSE value for each subset with varying proportions of White and Black participants. It was assumed that RMSE values would vary between individuals and therefore a random intercept was included and the following covariates were included: ratio (proportion of White and Black), race, age, sex, and training data sample size. (Categorical: Race, age, and sex; continuous: RMSE, ratio, and sample size) It was further assumed that the varying proportions of Black and White participants in the training data subsets would affect the performance and an interaction term was therefore added. The interaction term allowed for different slopes in the LMEM which allowed us to test the hypothesis

Surveillance error grids (SEG) were used to get a general idea of the clinical risk of the prediction models.[25] The error grids are originally developed for the regulatory bodies to asses clinical accuracy of blood glucose monitoring in post-surveillance before the wide spread use of CGM devices.[26] and the results should therefore be interpreted with caution when using CGM data. The SEG is a 600x600 grid assigned an individual risk score to each datapoint with 15 colored zones, each indicating severity of a monitoring error that is acted upon in regards to risk of hypoglycemia or hyperglycemia (Figure 3.7).[26] We decided to focus on the five colored zones regardless of the risk being hypo- or hypoglycemic, since one of the critiques of the SEG is that it is hard to interpret with all the zones and that the preferred number of zones are five and it is possible to only asses the output in five zones.[25] The five zones were "No", "Slight", "Moderate", "High", and "Extreme" errors according to the absolute error between the predicted and the actual value and the risk zone the error occurred in. Predictions outside of 0–33 mmol/L were changed to 0 or 33 mmol/L to keep within the upper and lower bounds of the surveillance error grid.

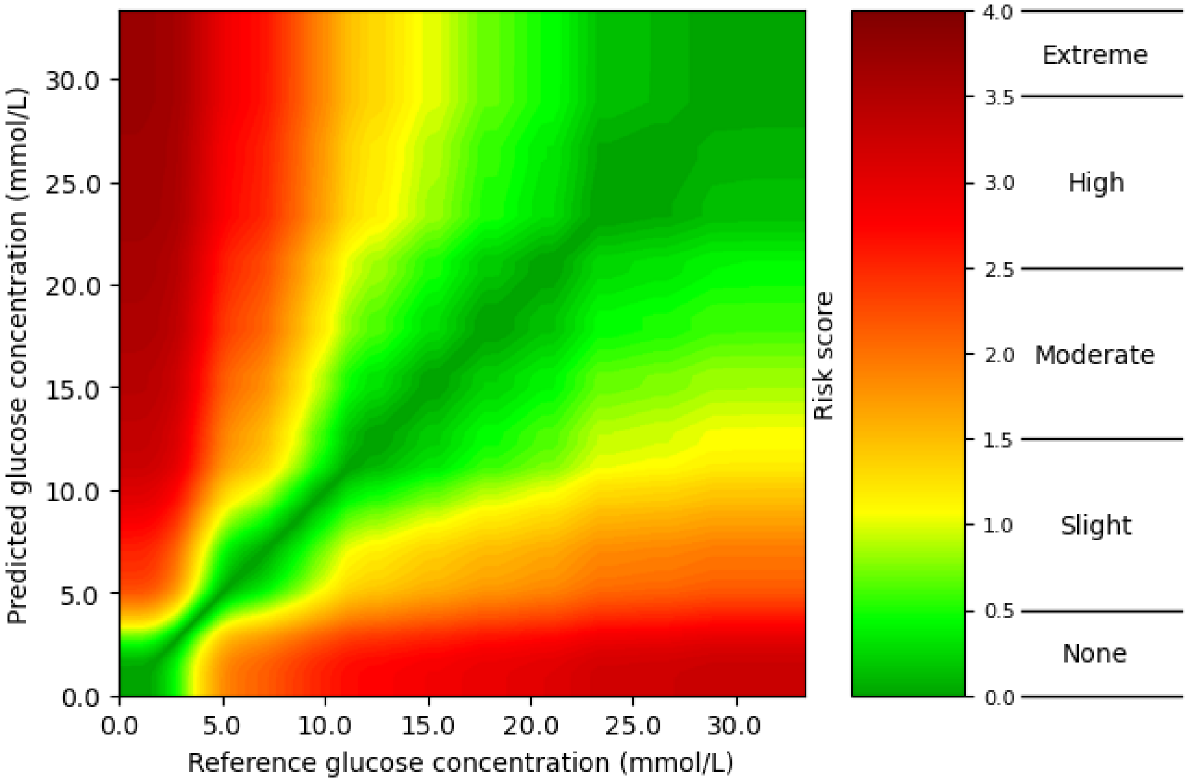


Figure 3.7.: An empty surveillance error grid

3.6. Bias in machine learning models (Study III)

- ☐ Fairness metrics
- ☐ Mitigation strategies -> Fine-tuning LSTM
- ☐ Regression analysis and coefficients?
- ☐ Evaluate bias when output is not binary (e.g., LMEM models)
- ☐ CGM specific error grids

4 Helenes Notes**

- ☒ Study I
- ☐ Study II
- ☐ NT-proBNP as a biomarker of cardiovascular risk
- ☐ Present CGM-derived metrics
- ☒ Study II

Results 5

5.1. metrics used in the papers - Study 1 - Scoping review

- ☐ CGM metrics and CVD definitions'
- ☐ Demographics
- ☐ Study designs / cohorts
- ☐ Diabetes types
- ☐ CGM metrics
- ☐ CVD risk factors
- ☐ Statistical methods used
- ☐ Check big and small letters in headers (uppercase/lowercase beginnings)

5.2. Study I: Results from a Scoping Review

5.2.1. Study Selection

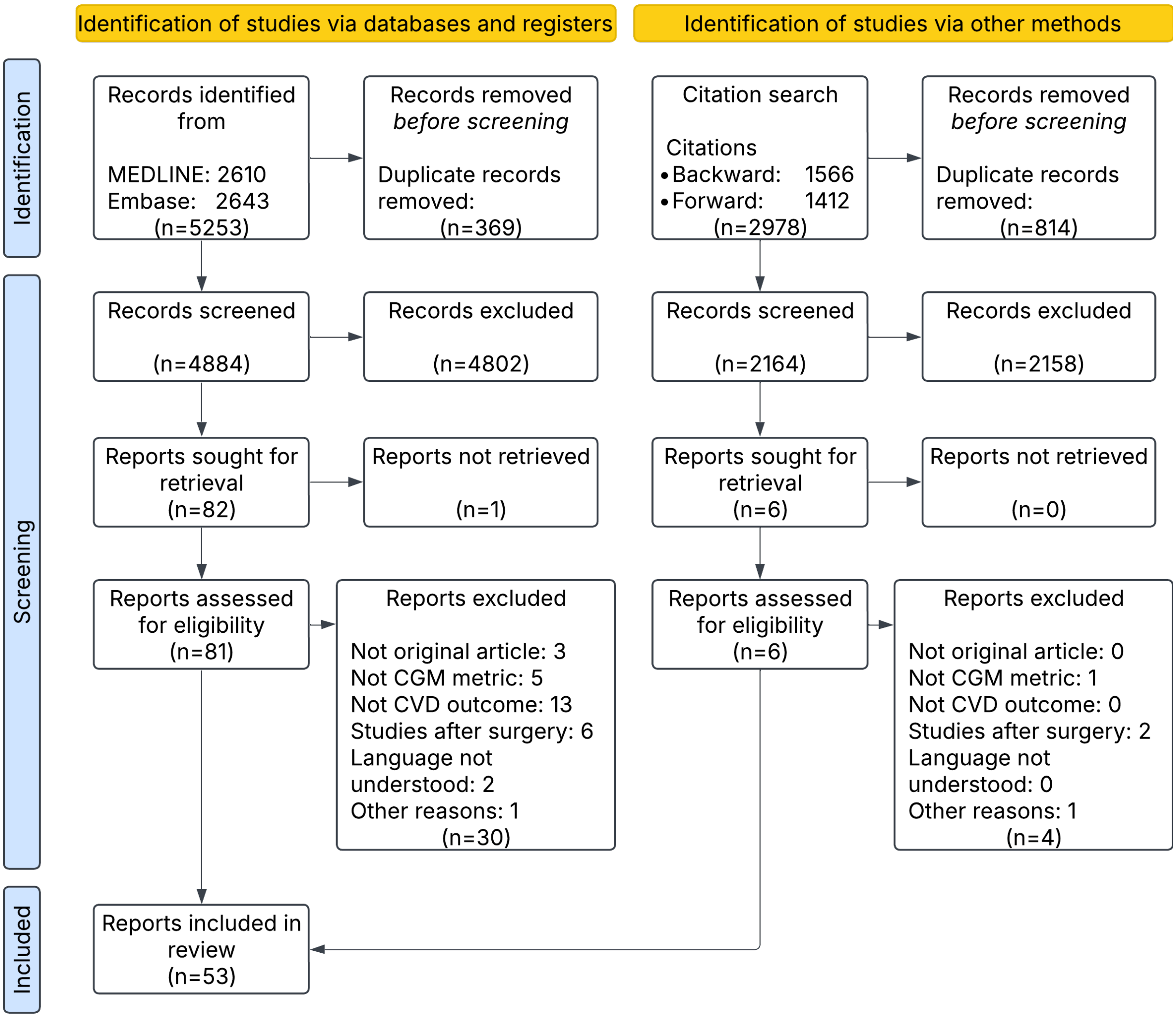
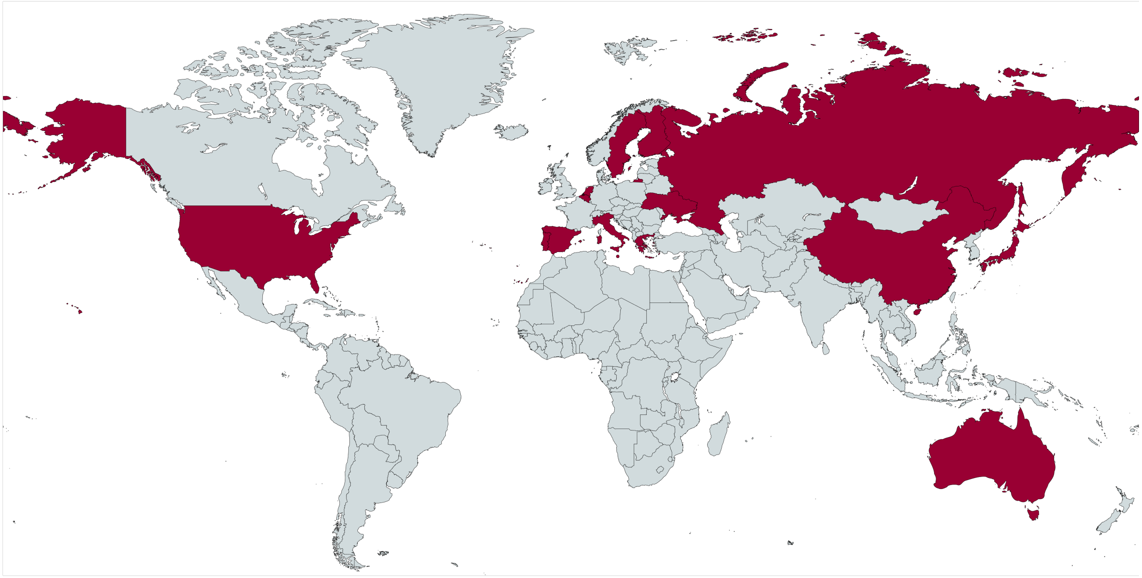


Figure 5.1.: Flow diagram of study selection

5.2.2. Study Populations

- [] Add mapchart.net to the referencelist and cite it in this figure text
- [] If timer permits it then change colorscale to viridis or AU blue ::: {#fig-prisma}



Geographical location of the studies (colored in red) were found based on the hospitals and centers where data was collected or by the first author's affiliation when the other option was not available. Illustration was created with mapchart.net

Geographic distribution of the studies :::

5.2.3. Study Designs and Demographics

- ☐ Check table is correct with the published article
- ☐ Add a raised letter to each abbreviation and include it in the table when relevant
- ☐ Add references to all citations

Table 5.1.: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Chen, 2020[†]	Longitudinal (prospective) and cross-sectional (retrospective)FU: in-hospital or within 3 months after discharge from hospital	T2D with CAD (Only male)BG control, n = 90BG fluctuation, n = 120	210	BG controls: 55.53 ± 7.30BG fluctuation: 56.41 ± 7.67	BG controls: 6.59 ± 2.30BG fluctuation group: 6.92 ± 2.25	nr	2 days
He, 2023	Longitudinal (prospective)FU: 1 year	T2D with kidney disease on hemodialysisHigh TIR, n = 12Low TIR, n = 15	27	High TIR: 66 [63, 73]Low TIR: 70 [64, 75]	High TIR: 2 [1.7, 10]Low TIR: 5 [0.75, 20]	High TIR: 43 [37, 51] mmol/mol (6.1 [5.5, 6.8]%)Low TIR: 66 [46, 70] mmol/mol (8.2 [6.4, 8.6]%)	14 days
Lu, 2021	Longitudinal (prospective)FU: until death occurred or 3–13 yearsMedian: 6.9 years	T2D, hospitalized	6225	mean age = 61.7 years	mean: 9.7 years	74.0 ± 24.0 mmol/mol (8.9 ± 2.2%)	72 hours

Table 5.1.: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Wei, 2019	Longitudinal (prospective)Median FU [IQR]: 31 months (22, 56)	T2D Divided into three groups:1) No hypo, n = 11732) Mild hypo (lvl1), n = 3233) Severe hypo (lvl3), n = 24	1520	No hypoglycemia: 58.59 ± 11.26Hypoglycemia: 62.27 ± 11.58	No hypoglycemia: 6.46 ± 6.00Hypoglycemia: 7.78 ± 7.37	No hypoglycemia: (8.19 ± 2.10%)Hypoglycemia: (7.73 ± 1.96%)	3 days
Bezerra, 2023	Cross-sectional	T1D	161	37.4 ± 13.4	17.7 ± 10.6	(7.5 ± 1.1%)	14 days
De Meulemeester, 2024†	Cross-sectional	T1D	808	44.8 ± 15.2	23.1 ± 13.6	63 ± 13 mmol/mol (7.9 ± 1.2%)	2 weeks
Deng, 2023	Cross-sectional	T2D	860	HP 1 carriers: 53.5 ± 13.3HP 2-2: 51.7 ± 14.6	HP 1 carriers: 8.6 ± 6.4HP 2-2: 8.6 ± 6.7	HP1 carriers: 73.0 ± 24.0 mmol/mol (8.8 ± 2.2%)Hp2-2: 72.0 ± 23.0 mmol/mol (8.7 ± 2.1)	3 days
El Malahi, 2022	Cross-sectional	T1D starting on sensor-augmented pump therapy	515	Mean ± SD [years]:42.2 ± 12.5	22.3 ± 11.6	60 ± 9.8 mmol/mol (7.6 ± 0.9 %)	2 weeks
Guo, 2021	Cross-sectional	T1D or T2D with atrial fibrillationWith Stroke, n = 48Without Stroke, n = 462	510	Stroke: 70.3 ± 12.1No stroke: 68.1 ± 9.4	nr	No Stroke: 7.4 ± 2.1Stroke: 8.2 ± 1.7	72 hour

Table 5.1.: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Li, 2020	Cross-sectional	T2D with LEAD, n = 179T2D without LEAD, n = 157	336	Without LEAD: 55.94 ± 12.45With LEAD: 65.56 ± 11.99	Without LEAD: 6.92 ± 3.54With LEAD: 10.32 ± 4.14	Without LEAD: 7.85 ± 1.41%With LEAD: 8.97 ± 1.63%	72 hours
Magri, 2018†	Cross-sectional	T2D	121	64 [57, 68]	3 [2, 5]	median: 45 mmol/mol (6.8%)	72 hours
Mi, 2012	Cross-sectional	T2D with chest painWithout CAD, n = 202With CAD, n = 84	286	Without CAD: 62.8 ± 8.7With CAD: 66.6 ± 9.2	nr	Non-CAD: 7.51 ± 0.80%CAD: 7.75 ± 0.92%	72 hours (Only intermediate 48 hours were used)
Sheng, 2023	Cross-sectional	T2D, hospitalized	545	mean age of (61.22 ± 11.21) years	nr	8.51 ± 1.85%	7-14 days
Su, 2011	Cross-sectional	T2D with chest painWithout CAD, n = 92With CAD, n = 252	344	Without CAD: 61 ± 9With CAD: 65 ± 9	Without CAD: 4.8 ± 5.7With CAD: 6.5 ± 6.4	non-CAD: 7.5 ± 1.4%CAD: 7.6 ± 1.5%	72 hours (only 48 hours used)
Watanabe, 2017	Cross-sectional	Prediabetes, hospitalized	28	64.3 ± 12.8	nr	5.41 ± 0.35%	72 hours (Only used the middle 48 hours)

Table 5.1.: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Zhang, 2013	Cross-sectional	T2D with cardiovascular complications-Group A: healthy individualsGroup B: T2D without cardiovascular complications-Group C: T2D with cardiovascular complications	92	Group A: 56.3 ± 6.1Group B: 56.1 ± 6.6Group C: 61.7 ± 7.2	nr	Group A: 5.3 ± 0.3%Group B: 6.6 ± 1.2%Group C: 7.5 ± 1.4%	72 hours

† This study appears both in the clinical and subclinical disease outcome tables due to investigating multiple CVD outcomes.

Note: Age is reported in years either as an interval, mean ± SD, or median (IQR). Diabetes duration values originally reported in months were converted to years for consistency (months ÷ 12). *Abbreviations:* CGM, continuous glucose monitoring; FU, follow-up; T1D, type 1 diabetes; T2D, type 2 diabetes; CAD, coronary artery disease; hypo, hypoglycemia; LEAD, lower extremity arterial disease; IQR, interquartile range; nr, not reported.

5.2.4. Outcomes from the Studies in the Scoping Review

- ☐ Check table is correct
- ☐ Add references to all citations
- ☐ Correct the raised letters/dagger is the same between the two tables
- ☐ Delete the multimedia part in the footnote of the table

Table 5.2.: Main findings of the included studies on clinical cardiovascular outcomes. {tbl-colwidths='[40, 15, 15, 15, 15]'}

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	*P* value for the least adjusted findings	Most adjusted findings	*P* value for the most adjusted findings
Cardiovascular mortality				
Lu et al [21], 2021				
TIR [c] $> 85\%$	HR 1.00	$< .001$ (trend)	HR 1.00	.02 (trend)
TIR $71\% - 85\%$	HR 1.43 (0.95–2.14)	$< .001$ (trend)	HR 1.35 (0.90–2.04)	.02 (trend)
TIR $51\% - 70\%$	HR 1.66 (1.12–2.45)	$< .001$ (trend)	HR 1.47 (0.99–2.19)	.02 (trend)
TIR $\geq 50\%$	HR 2.15 (1.47–3.13)	$< .001$ (trend)	HR 1.85 (1.25–2.72)	.02 (trend)
TIR as a continuous variable (each 10% decrease)	HR 1.08 (1.03–1.13)	— [d]	HR 1.05 (1.00–1.11)	—
Wei et al [46][e], 2019				
Hypoglycemic events	HR 2.033 (1.211–3.413)	—	HR 2.642 (1.398–4.994)	—
Major adverse cardiovascular events				
He et al [45], 2023				
Blood glucose risk index	HR 0.97 (0.85–1.10)	.61	HR 0.98 (0.85–1.13)	.75
Low blood glucose index	HR 2.37 (1.16–4.83)	.02	HR 2.73 (1.21–6.16)	.02
High blood glucose index	HR 0.94 (0.81–1.08)	.38	HR 0.94 (0.81–1.09)	.44
Average of daily risk range	HR 1.00 (0.93–1.07)	$> .99$	HR 1.01 (0.93–1.09)	.80
GMI [f]	HR 0.98 (0.91–1.06)	.65	HR 0.99 (0.91–1.07)	.78
M-value	HR 0.98 (0.91–1.05)	.54	HR 0.98 (0.91–1.06)	.64
Wei et al [46][e], 2019				
Hypoglycemic events	HR 1.501 (1.207–1.866)	—	HR 1.615 (1.239–2.106)	$< .001$
Macrovascular complications				
De Meulemeester et al [48][e,g], 2024				
TIR	OR 0.939 (0.829–1.063)	$> .05$	OR 0.896 (0.738–1.087)	$> .05$
TITR [h]	OR 0.901 (0.775–1.047)	$> .05$	OR 0.933 (0.745–1.169)	$> .05$

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	P value for the least adjusted findings	Most adjusted findings	P value for the most adjusted findings
Bezerra et al [47], 2023				
TIR	OR 0.66 (0.46–0.93)	.02	OR 0.68 (0.39–1.16)	.15
Time below 54 mg/dL	OR 1.10 (0.88–1.38)	.39	OR 0.92 (0.62–1.34)	.65
TBR [i]	OR 0.93 (0.80–1.09)	.39	OR 0.77 (0.54–1.11)	.17
TAR [j]	OR 1.04 (1.01–1.08)	.01	OR 1.04 (0.99–1.10)	.08
Time above 250 mg/dL	OR 1.04 (1.00–1.08)	.03	OR 1.03 (0.97–1.09)	.29
CV [k]	OR 1.03 (0.95–1.11)	.52	OR 0.92 (0.81–1.06)	.25
GMI	OR 2.17 (1.14–4.11)	.02	OR 2.03 (0.77–5.37)	.15
Deng et al [49], 2023				
%CV tertile 1 (Hp1; reference)	OR 1.000	.07 (interaction)	OR 1.000	.008 (interaction)
%CV tertile 1 (Hp2-2; reference)	OR 1.000	.07 (interaction)	OR 1.000	.008 (interaction)
%CV tertile 2 (Hp1)	OR 1.483 (0.907–2.423)	.12	OR 1.048 (0.528–2.078)	.89
%CV tertile 2 (Hp2-2)	OR 1.399 (0.829–2.358)	.21	OR 0.659 (0.296–1.466)	.31
%CV tertile 3 (Hp1)	OR 2.347 (1.393–3.957)	.001	OR 2.461 (1.183–5.121)	.02
%CV tertile 3 (Hp2-2)	OR 1.217 (0.731–2.027)	.45	OR 0.540 (0.245–1.191)	.13
El Malahi et al [50], 2022				
TIR	—	—	—	> .05
SD	—	—	—	> .05
CV	—	—	—	> .05
Magri et al [27][g], 2018				
TBR	—	—	OR 1.12 (1.014–1.228)	.02
Lowest BG [l] value	—	—	—	—
Area under the TBR curve	—	—	—	—
Coronary artery disease				
De Meulemeester et al [48][e,g], 2024				
TITR	OR 1.039 (0.812–1.330)	> .05	OR 1.255 (0.874–1.803)	> .05

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	P value for the least adjusted findings	Most adjusted findings	P value for the most adjusted findings
TIR	OR 1.072 (0.866–1.328)	$> .05$	OR 1.164 (0.844–1.607)	$> .05$
Sheng et al [53], 2023				
TIR $< 20\%$	—	—	OR 2.143 (1.554–3.287)	—
TIR $20\% - 40\%$	—	—	OR 1.049 (0.945–2.022)	—
TIR $40\% - 60\%$	—	—	OR 0.854 (0.495–1.473)	—
TIR $60\% - 80\%$	—	—	OR 0.617 (0.423–1.312)	—
TIR $> 80\%$	—	—	OR 0.470 (0.143–1.545)	—
Chen et al [40][e,g], 2020				
Controls with SD < 1.40 mmol/L, MAGE [m] < 3.90 mmol/L, LAGE [n] < 4.40 mmol/L, MODD [o] < 0.83 mmol/L versus high BG fluctuations (myocardial Infarction)	—	—	$\chi^2 = 5.797$	.02
Controls with SD < 1.40 mmol/L, MAGE < 3.90 mmol/L, LAGE < 4.40 mmol/L, MODD < 0.83 mmol/L versus high BG fluctuations (angina pectoris)	—	—	$\chi^2 = 7.490$	.006
Wei et al [46][e], 2019				
Hypoglycemic events (myocardial Infarction)	HR 1.901 (1.067–3.389)	—	HR 1.549 (0.768–3.124)	.03
Hypoglycemic events (unstable angina pectoris)	HR 1.226 (0.857–1.753)	—	HR 1.218 (0.794–1.869)	.30
Shu-Hua et al [43][e], 2012				
MAGE level (≥ 3.4 mmol/L)	—	—	OR 2.286 (1.176–4.446)	.02

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	P value for the least adjusted findings	Most adjusted findings	P value for the most adjusted findings
Su et al [28][e], 2011				
MAGE $\geq$ 3.4 mmol/L	—	—	OR 2.612 (1.423–4.831)	.002
MAGE	—	—	AUC 0.618 (0.555–0.680)	.001
Gensini score				
Chen et al [40][e,g], 2020				
Controls with SD < 1.40 mmol/L, MAGE < 3.90 mmol/L, LAGE < 4.40 mmol/L, MODD < 0.83 mmol/L versus high BG fluctuations	—	—	$t = 6.210$	< .001
Watanabe et al [54][e], 2017				
MAGE	—	—	$r = 0.742$	< .001
Shu-Hua et al [43][e], 2012				
MAGE	—	—	Unstandardized coefficient $\beta = 4.817$; SE = 1.164; standardized coefficient $\beta = 0.170$; $t = 2.984$	.003
Su et al [28][e], 2011				
MAGE	—	—	Unstandardized $\beta = 7.010$; SE = 1.466; standardized $\beta = 0.237$; $t = 4.783$	< .001
Syntax score				
Watanabe et al [54][e], 2017				
MAGE	—	—	$r = 0.776$	< .001
Zhang et al [30][e], 2013				
MAGE	—	—	$r = 0.518$	.01
BG fluctuations from 00:00 to 03:00	—	—	$r = -0.442$	.04

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	P value for the least adjusted findings	Most adjusted findings	P value for the most adjusted findings
BG fluctuations from 03:00 to 06:00	—	—	$r = -0.208$	.34
BG fluctuations from 06:00 to 08:00	—	—	$r = 0.678$	< .001
BG fluctuations from 08:00 to 11:00	—	—	$r = 0.115$	.60
BG fluctuations from 11:00 to 13:00	—	—	$r = 0.523$	.01
BG fluctuations from 13:00 to 17:00	—	—	$r = 0.257$	.24
BG fluctuations from 17:00 to 19:00	—	—	$r = 0.358$	.09
BG fluctuations from 19:00 to 24:00	—	—	$r = -0.018$	.93
Stroke				
De Meulemeester et al [48][e,g], 2024				
TITR	OR 0.651 (0.470–0.902)	< .05	OR 0.546 (0.347–0.858)	< .01
TIR	OR 0.749 (0.588–0.955)	< .05	OR 0.617 (0.440–0.866)	< .01
Guo et al [51], 2021				
TIR: Q1 ($\leq 46\%$; reference)	OR 1.00	< .001	OR 1.00	< .001
TIR: Q2 ($46\% - 65\%$)	OR 0.86 (0.72–0.95)	< .001	OR 0.80 (0.68–0.92)	< .001
TIR: Q3 ($65\% - 81\%$)	OR 0.71 (0.61–0.81)	< .001	OR 0.64 (0.53–0.79)	< .001
TIR: Q4 ($> 81\%$)	OR 0.66 (0.58–0.80)	< .001	OR 0.59 (0.50–0.74)	< .001
TIR (per 10% increase)	OR 0.93 (0.85–0.98)	.008	OR 0.89 (0.82–0.95)	.001
Wei et al [46][e], 2019				
Hypoglycemic events	HR 1.691 (1.144–2.499)	—	HR 1.813 (1.110–2.960)	.06
Peripheral artery disease				

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	P value for the least adjusted findings	Most adjusted findings	P value for the most adjusted findings
De Meulemeester et al [48][e,g], 2024				
TITR	OR 0.680 (0.426–1.085)	> .05	OR 0.807 (0.382–1.703)	> .05
TIR	OR 0.736 (0.520–1.042)	> .05	OR 0.811 (0.470–1.398)	> .05
Lower extremity arterial disease				
Li et al [52], 2020				
TIR	OR 0.979 (0.968–0.991)	< .001	OR 0.979 (0.965–0.992)	.002
CV	OR 1.040 (1.003–1.078)	.04	OR 1.038 (0.996–1.081)	.08
SD	OR 1.325 (1.038–1.691)	.02	OR 1.158 (0.824–1.627)	.40
TIR-without LEAD [p] (1)	OR 1.00	—	OR 1.00	—
TIR-mild LEAD (1)	OR 0.98 (0.97–1.00)	.14	OR 0.99 (0.97–1.01)	.25
TIR-moderate LEAD (1)	OR 0.97 (0.95–0.99)	.007	OR 0.97 (0.95–0.99)	.01
TIR-without severe LEAD (1)	OR 0.96 (0.94–0.98)	.002	OR 0.96 (0.94–0.98)	.003
CV-without LEAD	—	—	OR 1.00	—
CV-mild LEAD	—	—	OR 1.03 (0.98–1.07)	.28
CV-moderate LEAD	—	—	OR 1.02 (0.96–1.09)	.48
CV-without severe LEAD	—	—	OR 1.02 (0.95–1.09)	.60
TIR-without LEAD (2)	—	—	OR 1.00	—
TIR-mild LEAD (2)	—	—	OR 0.97 (0.96–1.08)	.06
TIR-moderate LEAD (2)	—	—	OR 0.98 (0.95–0.99)	.01
TIR-without severe LEAD (2)	—	—	OR 0.97 (0.95–0.99)	.02
SD-without LEAD	—	—	OR 1.00	—
SD-mild LEAD	—	—	OR 0.88 (0.47–1.64)	.69
SD-moderate LEAD	—	—	OR 1.28 (0.58–3.07)	.58
SD-without severe LEAD	—	—	OR 1.52 (0.92–2.41)	.10

[a] The full table with adjustments is provided in **Multimedia Appendix 10**. Further elaboration on the adjusted variables can be found in **Multimedia Appendices 7 and 8**.
[b] All hazard ratios (HRs), odds ratios (ORs), and areas under the curve (AUCs) are reported as follows: point estimate (95% CI).
[c] TIR: time in range.
[d] Not applicable or not available/not reported.
[e] This study appears multiple times as it investigated multiple cardiovascular disease outcomes.
[f] GMI: glucose management indicator.
[g] This study appears in both the clinical and subclinical disease outcome tables owing to the investigation of multiple cardiovascular disease outcomes.
[h] TITR: time in tight range.
[i] TBR: time below range.
[j] TAR: time above range.
[k] CV: coefficient of variation.
[l] BG: blood glucose.
[m] MAGE: mean amplitude of glycemic excursions.
[n] LAGE: largest amplitude of glycemic excursions.
[o] MODD: mean of daily differences.
[p] LEAD: lower extremity arterial disease.

Table 5.3.: Summary of established clinical and subclinical outcomes and corresponding abbreviations identified in the literature.

Clinical	Subclinical
• CVD = Cardiovascular disease - Macrovascular complications - Diabetic macroangiopathy • CAD = Coronary artery disease - CHD = Coronary heart disease - MI = Myocardial infarction - Heart attack - Heart failure - (Unstable) angina pectoris • Cardiovascular events - Ishemic heart disease (IHD) • LEAD = Lower extremity artery disease • MACE = Major adverse cardiovascular events • PAD = Peripheral artery disease • Stroke = Stroke, Cerebrovascular disease, or Cerebrovascular accident	• ABI = Ankle–brachial index • BP = Blood pressure • Carotid artery distensibility = Carotid artery distensibility or carotid distensibility coefficient • CCA = Common carotid artery • CIMT = Carotid artery intima-media thickness • DBP = Diastolic blood pressure • FMD = Flow-mediated dilation of the brachial artery • GSM = Gray-scale median of the carotid arteries • HRV = Heart rate variability • IMT = Intima-media thickness • PWV = Pulse wave velocity - ba-PWV = Brachial-ankle pulse wave velocity - cr-PWV = Carotid-radial pulse wave velocity - cf-PWV = Carotid-femoral pulse wave velocity - Aortic PWV = Aortic pulse wave velocity • SBP = Systolic blood pressure

5.2.5.

5.3. Baseline characteristics table of the datasets

Table 5.4.: Baseline table for CANCAN data and NT-proBNP groups.

Variable	CANCAN, total	NT-proBNP 125 pg/mL	NT-proBNP > 125 pg/mL
N	159	92	67
Age (years)	65 [56, 71]	60 [53, 68]	68 [64, 76]
Sex			
– Male	99 (62%)	55 (60%)	44 (66%)
– Female	60 (38%)	37 (40%)	23 (34%)
Smoking status			
– Never or former	135 (85%)	77 (84%)	58 (87%)
– Current	24 (15%)	15 (16%)	9 (13%)
Diabetes duration (years)	16 [11, 24]	14 [9, 22]	19 [14, 25]
SCORE2-Diabetes (%)	14.8 [9.1, 20.9]	11.9 [7.8, 17.2]	19.0 [13.4, 25.4]
HbA1c (mmol/mol)	64 [55, 80]	66 [56, 86]	60 [54, 72]
Systolic blood pressure (mmHg)	132 [123, 143]	132 [123, 143]	131 [122, 142]
HDL (mmol/L)	1.0 [0.9, 1.2]	1.0 [0.9, 1.2]	1.0 [0.9, 1.3]
Total cholesterol (mmol/L)	3.9 [3.2, 4.8]	4.1 [3.5, 4.9]	3.6 [3.0, 4.2]
eGFR (mL/min/1.73 m^2)	57 [39, 74]	68 [52, 82]	48 [27, 63]
CGM-derived metrics			
– CGM days (n/participant)	13 [10, 13]	13 [9, 13]	13 [10, 13]

Variable	CANCAN, total	NT-proBNP 125 pg/mL	NT-proBNP > 125 pg/mL
– Glucose management indicator (%)	7.3 [6.8, 8.4]	7.5 [6.8, 8.6]	7.2 [6.7, 8.0]
– Time in range (%)	59.6 [31.9, 78.9]	58.2 [20.2, 78.6]	62.1 [40.8, 81.0]
– Time below range (%)	0.0 [0.0, 1.5]	0.0 [0.0, 0.8]	0.2 [0.0, 2.0]
– Time above range (%)	36.6 [17.6, 68.1]	41.8 [19.8, 79.5]	33.1 [14.1, 58.2]
– Coefficient of variation (%)	27.3 [22.5, 31.9]	25.3 [21.2, 30.3]	28.3[24.8, 34.2]
– Mean amplitude of glycemic excursions (mmol/L)	6.3 [5.0, 7.9]	6.0 [4.9, 7.7]	6.7 [5.1, 8.5]

Categorical values are given by number (percentage), and numerical values are given by median [interquartile range].

Table 5.5.: Baseline table for T1DX data and racial groups.

Characteristics	Total	White Cohort	Black Cohort
N	205	101	104
Race (White)	101 (49%)	101 (100%)	0 (0%)
Sex (male)	88 (43%)	46 (46%)	42 (40%)
Age group (child, < 18 yrs)	82 (40%)	49 (49%)	33 (32%)
Age (years)	28 [15, 47]	22 [13, 46]	32 [16, 47]
BMI (kg/m^2)	25.1 [21.7, 29.2]	24.0 [20.7, 28.6]	26.7 [22.3, 30.0]
Age at T1D diagnosis	11 [7, 22]	9 [6, 16]	14 [8, 25]
HbA1c (mmol/mol)	67 [57, 78]	63 [55, 73]	72.0 [60, 85]
CGM duration w/o missing data (weeks/participant)	9.1 [7.2, 10.3]	9.5 [7.7, 10.7]	8.6 [6.9, 9.9]
Missing CGM values (weeks/per participant)	3.3 [1.6, 5.1]	2.6 [1.3, 4.5]	4.1 [1.8, 5.6]
CGM-derived metrics			
– CGM days (n/participant)	58 [46, 69]	63 [47, 71]	56 [45, 66]
– Glucose management indicator (%)	7.5 [7.0, 8.2]	7.4 [6.9, 7.9]	7.7 [7.0, 8.5]
– Time in range (%/per participant)	47 [35, 57]	49 [39, 60]	42 [31, 54]
– Time above range (%/per participant)	45 [30, 56]	42 [28, 53]	48 [32, 60]
– Time below range (%/per participant)	8 [4, 14]	6 [4, 10]	11 [5, 15]
– Coefficient of variation (%)	43.3 [39.6, 48.6]	42.2 [38.6, 46.5]	46.7 [40.4, 51.9]

Characteristics	Total	White Cohort	Black Cohort
– Mean amplitude of glycemic excursions (mmol/L)	9.9 [8.2, 12.1]	9.6 [8.1, 11.2]	10.6 [8.5, 13.0]

Categorical values are given by number (percentage), and numerical values are given by median [interquartile range].

5.4. Study II: Results from a Foundation Model Embedding Analysis

5.4.1. Representation learning of CGM data (Study 2)

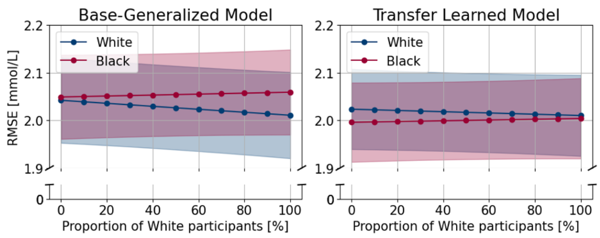
5.4.2. Prediction of NT-proBNP (Study 2)

- Prediction NT-proBNP with different baseline models
- Predicting NT-proBNP using only age
- PRedicting NT-proBNP using only CGM metrics
- Predicting NT-proBNP using SCORE2-Diabetes inputs and score together with other CGM data -> Supplementary material of paper 2

5.5. Study III: Results from an Algorithmic Fairness Study

5.5.1. Bias in Machine Learning models (Study 3)

- Baseline characteristics of the T1DX cohort stratified by race status
- Fairness metrics of the LMEM model before and after fine-tuning



Note: The y-axis starts at 1.9 mmol/L and not 0

Figure 5.2.: Estimated root mean squared errors (RMSE) across 11 different proportions of White and Black participants in the dataset.

Ratio	LOCF	Base-Individual	Base-Generalized	Transfer Learned
Model performance (RMSE) for White participants				
0			2.04 [1.95, 2.13]	2.02 [1.94, 2.11]
10			2.04 [1.95, 2.13]	2.02 [1.94, 2.11]
20			2.04 [1.95, 2.12]	2.02 [1.94, 2.10]

Ratio	LOCF	Base-Individual	Base-Generalized	Transfer Learned
30			2.03 [1.94, 2.12]	2.02 [1.94, 2.10]
40	2.45 [2.32, 2.57]	2.54 [2.38, 2.70]	2.03 [1.94, 2.12]	2.02 [1.94, 2.10]
50	(independent of ratio)	(independent of ratio)	2.03 [1.94, 2.11]	2.02 [1.93, 2.10]
60			2.02 [1.94, 2.11]	2.02 [1.93, 2.10]
70			2.02 [1.93, 2.11]	2.01 [1.93, 2.10]
80			2.02 [1.93, 2.11]	2.01 [1.93, 2.10]
90			2.01 [1.92, 2.10]	2.01 [1.93, 2.10]
100			2.01 [1.92, 2.10]	2.01 [1.92, 2.10]
Model performance (RMSE) for Black participants				
0			2.05 [1.96, 2.14]	2.00 [1.91, 2.08]
10			2.05 [1.96, 2.14]	2.00 [1.91, 2.08]
20			2.05 [1.96, 2.14]	2.00 [1.92, 2.08]
30			2.05 [1.97, 2.14]	2.00 [1.92, 2.08]
40	2.41 [2.28, 2.53]	2.63 [2.47, 2.79]	2.05 [1.97, 2.14]	2.00 [1.92, 2.08]
50	(independent of ratio)	(independent of ratio)	2.05 [1.97, 2.14]	2.00 [1.92, 2.08]
60			2.06 [1.97, 2.14]	2.00 [1.92, 2.08]
70			2.06 [1.97, 2.14]	2.00 [1.92, 2.08]
80			2.06 [1.97, 2.14]	2.00 [1.92, 2.09]
90			2.06 [1.97, 2.15]	2.00 [1.92, 2.09]
100			2.06 [1.97, 2.15]	2.00 [1.92, 2.09]

Values that vary by proportion are estimated using linear mixed-effects models. These models were adjusted to the population's age (40% children) and sex (57% women) distribution. Abbreviations: LOCF = last observation carried forward.

Root mean squared errors with 95% CIs for the different models.

Predictor	Coefficient	95% CI	P-value
Intercept	1.893	[1.762, 2.024]	<0.001
Race			
White	Reference		
Black	0.007	[-0.118, 0.132]	0.91
Age			
Adult	Reference		
Child	0.482	[0.355, 0.610]	<0.001
Sex			
Male	Reference		
Female	-0.076	[-0.201, 0.048]	0.23
Training size	0.019	[-0.055, 0.093]	0.62
Base-Generalized (per 100,000 samples)*			
Ratio (per 10%)	-0.003	[-0.007, 0.001]	0.12
Ratio · Race	0.004	[0.001, 0.008]	0.02

*Training size was centered at 510,000 as the training data had a mean of 514,293 samples.

Model coefficients from linear mixed effects models for root mean squared errors of the Base-Generalized models.

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Predictor	Coefficient	95% CI	P-value
Intercept	1.890	[1.768, 2.015]	<0.001
Race			
White	Reference		
Black	-0.028	[-0.147, 0.092]	0.65
Age			
Adult	Reference		
Child	0.411	[0.290, 0.533]	<0.001
Sex			
Male	Reference		
Female	-0.057	[-0.173, 0.060]	0.34
Training size Base-Generalized (per 100,000 samples)*	0.011	[-0.060, 0.081]	0.77
Training size tl (per 1,000 samples)**	-0.111	[-0.161, -0.061]	<0.001
Ratio (per 10 units)	-0.001	[-0.003, 0.001]	0.46
Ratio · Race	0.002	[-0.001, 0.005]	0.20

*Training size was centered at 510,000 as the training data had a mean of 514,293 samples.
**Training size for the transfer learned approach was centered at 4,400 as the training data for fine-tuning had a mean of 4,371 samples.

Model coefficients from linear mixed effects models for root mean squared errors of the Transfer Learned models.

6 Discussion

7 Main findings

- Scoping review highlighted a chaotic landscape of wildly different study designs in the literature
- Mention other reviews, Yapanis + the other one that I forgot
- Representation learning of CGM data can capture complex patterns in glucose dynamics that are not captured by traditional CGM metrics
- Machine learning models can predict NT-proBNP levels using CGM data, however the performance of these models is modest and there is still room for improvement
- Bias is a concern in machine learning models, and fine-tuning on a subset of the data can help mitigate bias as measured by fairness metrics
- Grouped fairness is easier to evaluate and mitigate than individual fairness, but it is still an important step in ensuring that machine learning models are equitable and do not perpetuate existing biases in healthcare.

e learning vs traditional statistical methods 8

9 something about data and cross-sectional v

- Mention the collins protocol that creates a study focused on answer this question
- Mention the CANCAN dataset and the weird trend with people being better of the more comobidities

9.1. Study 3 Method choices

PREDICTION HORIZON Many studies in the literature tests a variety of different prediction horizons but it seems to always predict best, the shorter the prediction horizon is. Most often sensors do also have a higher resolution then 15 minuts (3-5 minutes resolution)

- [ ] In the discussion I could mention LSTM vs CNN vs MLP and talk about the difference in model doesn't make that big of a difference.

Clinical implications 10

11 Strengths and limitations

12 Conclusions and future perspectives

13 Conclusion

- CGM data has the potential to provide valuable insights into cardiovascular risk in diabetes, but there are still many challenges to overcome in terms of study design, data quality, and machine learning methods.

Perspectives 14

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English summary

Background: Type 2 diabetes (T2D) is an increasing burden on healthcare services due to changes in demography and lifestyle. Migrants are particularly vulnerable to T2D, having a higher prevalence of the disease compared to native populations in western Europe, with some evidence suggesting a higher risk of complications and mortality. Evidence on disparities in care that may contribute to this excess risk in migrants is sparse, but register-based studies are well-suited to explore this in a Danish context. However, these studies are vulnerable to biases due to the lack of a validated method to identify T2D cases in the general population.

Aims: This dissertation aimed to develop and validate a register-based classification of type 1 diabetes (T1D) and T2D, and utilise it to explore specific areas of monitoring, biomarker levels, and pharmacological treatment, where disparities in T2D care between migrant groups and native Danes may contribute to excess complication risk in migrants.

Methods: Three cross-sectional register-based studies were performed. Study I validated two register-based classifications of diabetes - the *Open-Source Diabetes Classifier* (OSDC), and the *Register of Selected Chronic Diseases* (RSCD) - in a population of survey respondents from Central Denmark Region (*Health In Central Denmark*), while the two subsequent studies examined T2D prevalence and care in nationwide populations, using native Danes as the reference group. Study II computed relative risk (RR) of prevalent T2D and non-fulfilment of eleven indicators corresponding to guideline-recommendations. Study III computed RR of using glucose-lowering drug (GLD) combination therapy, and RR of using each type of GLD.

Results: In study I, both register-based diabetes classifiers identified valid populations of T1D and T2D in a general population (positive predictive values from 87.5% to 94.4%, negative predictive values above 98.9%), although sensitivity in the OSDC was substantially higher compared to the RSCD in both T1D (77.3% vs. 70.0%) and in T2D (94.4% vs. 90.5%). Neither algorithm was able to accurately classify diabetes type in individuals with T1D onset after age 40, nor T2D onset before age 40. In study II, prevalence of T2D was higher in all migrants excluding the Europe-group (fully-adjusted RR from 0.96 [95% confidence interval: 0.94-0.99] (Europe group) to 4.04 [3.89-4.20] (Sri Lanka group)). Non-fulfilment of recommendations for care was common regardless of origin: in eight of the eleven indicators of care studied >25% of native Danes did not meet guideline-recommendations. Apart from monitoring in the Sri Lanka group, migrants were at similar or higher risk of non-fulfilment than native Danes across all indicators of monitoring (crude RR from 0.64 [0.51-0.80] (hemoglobin-A1c (HbA1c) monitoring, Sri Lanka group) to 1.62 [1.36-1.95] (HbA1c monitoring, Somalia group)), HbA1c control (crude RR from 1.27 [1.24-1.30] (Middle East group) to 1.46 [1.41-1.52] (Pakistan group)), and low-density lipoprotein cholesterol (LDL-C) control (crude RR from 1.08 [1.03-1.14] (Former Yugoslavia group) to 1.78 [1.67-1.90] (Somalia group)), while no overall disparities were observed for the presence of pharmacological treatment (crude RR from 0.61 [0.46-0.80] (GLD, Sri Lanka group) to 1.67 [1.34-2.09] (GLD, Somalia group)). In study III, migrants were less likely to use GLD combination therapy than native Danes (fully adjusted RR from 0.77 [0.71-0.85] (Somalia group) to 1.00 [0.97-1.04] (Former Yugoslavia group)). Migrants were more likely to use oral GLD types (fully adjusted RR (use of any oral GLD) from 0.99 [0.97-1.01] (Europe group) to 1.09 [1.06-1.11] (Sri Lanka group)), but less likely to use injection-based GLD types such as insulins

(fully adjusted RR from 0.66 [0.62-0.71] (Sri Lanka group) to 0.94 [0.89-0.99] (Europe group)) and, especially, glucagon-like peptide-1 receptor agonists (GLP1RA) (fully adjusted RR from 0.29 [0.22-0.39] (Somalia group) to 0.95 [0.89-1.01] (Europe group)).

Conclusion and perspectives This dissertation was able to accurately identify populations of T1D and T2D in register data from the general Danish population. Diabetes epidemiologists can rely on the OSDC and the RSCD to provide valid study populations for Danish register-based research, but researchers should be aware of limitations in cases with atypical age at onset and potential challenges due to evolving GLD indications in the future. This dissertation found room for improvement in T2D care in Denmark, and presented several disparities in T2D care between migrants and native Danes. These may contribute excess risk of complication and mortality in migrants, especially among migrants from Somalia, who received the poorest care of all groups overall. Overall, the large disparity in use of GLP1RA between migrants and native Danes may provide a target for future interventions to improve care and reduce complication risk among migrants with T2D.

Dansk resumé

Baggrund: Type 2 diabetes (T2D) er en stigende belastning på sundhedsvæsenet på grund af ændringer i demografi og livsstil. Indvandrere er særligt sårbare overfor T2D, idet de har en højere forekomst af sygdommen i forhold til indfødte befolkninger i Vesteuropa, ligesom nogle undersøgelser peger på en højere risiko for diabetiske komplikationer og død. Potentielle uligheder i behandlingen, der kan bidrage til den øgede risiko hos indvandrere med T2D, er dårligt belyst, men registerbaserede studier er velegnede til at undersøge dette i en dansk sammenhæng. Disse undersøgelser er imidlertid sårbare, idet der ikke foreligger nogen validerede metoder til at identificere T2D-tilfælde i den generelle danske befolkning.

Formål: Denne afhandling havde til formål at udvikle og validere en registerbaseret klassifikation af type 1 diabetes (T1D) og T2D, og anvende den til at undersøge specifikke områder i behandlingen af T2D - kontroller, biomarkør-niveauer og medicinsk behandling - hvor uligheder mellem indvandrergrupper og indfødte danskere kan bidrage til øget risiko for komplikationer hos indvandrere.

Metoder: Som led i afhandlingen blev tre tværsnitsstudier baseret på register-data gennemført. Studie I validerede to registerbaserede klassifikationer af diabetes - *Open-Source Diabetes Classifier* (OSDC) og *Register for Udvalgte Kroniske Sygdomme* (RUKS) - blandt spørgeskema-besvarelser fra personer i Region Midtjylland (*Helbred i Midt*), mens de to efterfølgende studier undersøgte forekomst og behandling af T2D i hele landet og sammenlignede indvandrergrupper med indfødte danskere. Studie II beregnede relativ risiko (RR) for forekomst af T2D og for manglende opfyldelse af elleve indikatorer for behandlingskvalitet svarende til anbefalinger i gældende kliniske retningslinjer. Studie III beregnede RR for brug af kombinationsbehandling med blodsukker-nedsættende lægemidler (GLD) og RR for brug af hver type GLD.

Resultater: I studie I identificerede begge registerbaserede diabetesklassifikationer gyldige populationer af T1D og T2D i en bred befolkning (positive prædiktive værdier fra 87,5% til 94,4%, negative prædiktive værdier over 98,9%), dog med en væsentlig højere sensitivitet i OSDC sammenlignet med RUKS både i T1D (77,3% vs. 70,0%) og i T2D (94,4% vs. 90,5%). Ingen af algoritmerne var i stand til nøjagtigt at klassificere diabetestype hos personer med T1D debut efter 40-årsalderen eller T2D debut før 40-årsalderen. I studie II var forekomsten af T2D højere i alle indvandrergrupper undtagen Europa-gruppen (fuldt justeret RR fra 0,96 [95% konfidensinterval: 0,94-0,99] (Europa-gruppen) til 4,04 [3,89-4,20] (Sri Lanka-gruppen)). Manglende opfyldelse af de vejledende behandlingsanbefalinger var hyppig - uanset oprindelse: i otte ud af elleve indikatorer opfyldte >25% af indfødte danskere ikke anbefalingerne. Fraset rutinemæssig kontrolundersøgelser i Sri Lanka-gruppen, havde indvandrergrupperne tilsvarende eller højere risiko end indfødte danskere for ikke at opfylde behandlingsanbefalinger på tværs af alle indikatorer for rutinemæssig kontrolundersøgelser (rå RR fra 0,64 [0,51-0,80] (hæmoglobin-A1c (HbA1c-undersøgelse, Sri Lanka-gruppen) til 1,62 [1,36-1,95] (HbA1c-undersøgelse, Somalia-gruppen)), blodsukkerniveau (rå RR fra 1,27 [1,24-1,30] (Mellemøsten-gruppen) til 1,46 [1,41-1,52] (Pakistan-gruppen)) og kolesterolniveau (rå RR fra 1,08 [1,03-1,14] (Eksjugoslavien-gruppen) til 1,78 [1,67-1,90] (Somalia-gruppen)), mens der ikke sås nogen gennemgående forskelle for tilstedeværelse af medicinsk behandling (rå RR fra 0,61 [0,46-0,80] (GLD, Sri Lanka-gruppen) til 1,67 [1,34-2,09] (GLD, Somalia-gruppen)). I studie III var indvandrere mindre tilbøjelige til at

få GLD kombinationsbehandling end indfødte danskere (fuldt justeret RR fra 0,77 [0,71-0,85] (Somalia-gruppen) til 1,00 [0,97-1,04] (Eksjugoslavien-gruppen)). Indvandrere var mere tilbøjelige til at benytte orale GLD typer (fuldt justeret RR (brug af ethvert oralt GLD) fra 0,99 [0,97-1,01] (Europa-gruppen) til 1,09 [1,06-1,11] (Sri Lanka-gruppen)), men mindre tilbøjelige til at benytte injektionsbaserede GLD typer så som insuliner (fuldt justeret RR fra 0,66 [0,62-0,71] (Sri Lanka-gruppen) til 0,94 [0,89-0,99] (Europa-gruppen)) og især glucagon-like peptide-1 receptor agonister (GLP1RA) (fuldt justeret RR fra 0,29 [0,22-0,39] (Somalia-gruppen) til 0,95 [0,89-1,01] (Europa-gruppen)).

Konklusion og perspektiver: Denne ph.d.-afhandling kunne præcist identificere befolkninger med T1D og T2D i den brede danske befolkning ud fra registerdata. Diabetesforskere kan anvende OSDC og RUKS til at danne gyldige studiepopulationer ud fra de danske registre, men bør være opmærksomme på deres begrænsninger, herunder blandt patienter med atypisk alder for sygdomsdebut samt potentielle udfordringer som følge af udviklingen i brugen af GLD i fremtiden. Afhandlingen fandt plads til forbedringer i behandlingen af T2D i Danmark, og viste flere uligheder mellem indvandrere og indfødte danskere i behandlingen af T2D, som kan medføre øget risiko for komplikationer og død blandt indvandrere - særligt blandt indvandrere fra Somalia, som havde den laveste behandlingskvalitet af alle grupper. Den store forskel i brugen af GLP1RA mellem indvandrere og indfødte danskere kan muligvis være et mål for fremtidige interventioner til at øge behandlingskvaliteten og mindske komplikationsrisikoen blandt indvandrere med T2D.

Supplementary validation analyses A

A.0.1. Validation of unspecified diabetes in the Health In Central Denmark survey and in the National Health Survey.

Validation analyses performed in the HICD study population from study I:

Validation analyses performed among respondents in the National Health Survey 2017 (individuals reporting diabetes as a current disease were considered diabetes cases).

Table A.1.: **Supplementary: Validation of uspecified diabetes in the HICD study population**

Table A.2.: The Open-Source Diabetes Classifier (OSDC)

OSDC	Survey: +DM	Survey: -DM	Total N
OSDC: +DM	2,512	221	2,733
OSDC: -DM	121	26,537	26,658
Total N	2,633	26,758	29,391
Sensitivity	0.954 (0.945, 0.962)		
Specificity	0.992 (0.991, 0.993)		
PPV	0.919 (0.908, 0.929)		
NPV	0.995 (0.995, 0.996)		

Table A.3.: The Register for Selected Chronic Diseases (RSCD)

RSCD	Survey: +DM	Survey: -DM	Total N
RSCD: +DM	2,422	122	2,544
RSCD: -DM	211	26,636	26,847
Total N	2,633	26,758	29,391
Sensitivity	0.920 (0.909, 0.930)		
Specificity	0.995 (0.995, 0.996)		
PPV	0.952 (0.943, 0.960)		
NPV	0.992 (0.991, 0.993)		

Table A.4.: **Supplementary: Validation of uspecified diabetes in the National Health Survey**

Table A.5.: The Open-Source Diabetes Classifier (OSDC)

OSDC	Survey: +DM	Survey: -DM	Total N
OSDC: +DM	1,630	258	1,888
OSDC: -DM	87	26,460	26,547
Total N	1,717	26,718	28,435
Sensitivity	0.95 (0.94, 0.96)		
Specificity	0.99 (0.99, 0.99)		
PPV	0.86 (0.85, 0.88)		
NPV	1.00 (1.00, 1.00)		

Table A.6.: The Register for Selected Chronic Diseases (RSCD)

RSCD	Survey: +DM	Survey: -DM	Total N
RSCD: +DM	1,546	184	1,730
RSCD: -DM	171	26,534	26,705
Total N	1,717	26,718	28,435
Sensitivity	0.90 (0.89, 0.91)		
Specificity	0.99 (0.99, 0.99)		
PPV	0.89 (0.88, 0.91)		
NPV	0.99 (0.99, 0.99)		

Full papers B

B.1. Paper I

B.2. Paper II

B.3. Paper III

C Addenda and errata

This chapter contains an overview of:

- Addenda: contents added since the submitted version of the disseration (commit db88939).
- Errata: corrections to errors discovered since submission of the dissertation.

C.1. Addenda

- 5 May 2023:
  - Study I paper published. List of studies updated with citation and link to doi.
- 8 August 2023:
  - Study III paper published. List of studies updated with citation and link to doi.

C.2. Errata

- 2 May 2023:
  - On lines 5-10, page 81, section 5.1.5., the following excerpt on the interpretation of **?@fig-age-inclusion-by-year** states:
  - "Before 2016, when HbA1c data was not available from all regions of Denmark, a spike of inclusions is seen for women with T2D included at age 40 years. This spike is absent in the subsequent years, but the distribution is slightly skewed with a plateau in the age range 40-45 years. This is likely a residual effect of women in this age range having been prematurely included in the preceding years."
    - This statement misinterprets some elements of the figure, which are addressed below:
    - The density plots are smoothed, and spikes in single years are less visible compared to a histogram of observed counts (due to statistical disclosure requirements, this data could not be visualised using histograms). While the spike in included OSDC T2D-cases among women age 40 is indeed much lower in the years 2016-2018 than in the previous years, it is still present.
    - Most likely, the spike is caused by women with onset of T2D before the age of 40 years during a period of time without HbA1c data. These women may have initiated metformin treatment before HbA1c data became available, and if they achieved stable euglycaemia using only metformin, they would not be be captured by the HbA1c inclusion criterion after the HbA1c data became available. Thus, their inclusion into the cohort would be delayed until the censoring of metformin purchases ceases at age 40 years, causing a spike at this age of inclusion. In the

post-2016 data, women living in the Southern Denmark Region (which did not supply HbA1c data prior to 2015, compared to the other regions, which supplied data from 2011 onwards) are likely the major source of this spike.